MIL-T-22053A(EC)
11 February 1969
SUPERSEDING
MIL-T-22053(SHIPS)
6 May 1959
(See 6.3)

MILITARY SPECIFICATION

TEST CHAMBER, RADIAC DETECTOR, TS-1189/PD

1 SCOPE

1.1 This is a general performance specification which covers a radiac detector test chamber TS-1189/PD for evaluating the accuracy of any 0 to 200 milliroentgen (mr) radiacmeter in the Navy system

2 APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-E-15090 — Enamel, Equipment, Light-Gray (Formula No 111)
MIL-B-15395 — Brazing Alloys, Silver
MIL-P-17555 — Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-M-19590 — Marking of Commodities and Containers to Indicate Radioactive Material

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following document forms a part of this specification to the extent specified herein Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply

DEPARTMENT OF TRANSPORTATION
CODE OF FEDERAL REGULATIONS
TITLE 49 - Transportation, Parts 0 to 190

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D C 20402.)

3 REQUIREMENTS

3.1 First article sample - Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.2)

3.2 General description - The radiacmeter test chamber is a portable equipment capable of evaluating the accuracy of any 0 to 200 mr and 0 to 1 R radiacmeter in the Navy system

3.3 Material - Materials shall be chosen with care from a standpoint of complete suitability for the purpose intended

3.3.1 Silver brazing alloy shall conform to grade III of MIL-B-15395

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3 4 Equipment composition - The calibration equipment shall consist of the following:

(a) Radiation source and container
(b) Cylinder
(c) Well
(d) Baseplate
(e) Carrying strap
(f) Lock

3 4 1 Radiation source - The radiation source shall be Cesium 137 and shall be contained in a ceramic carrier

* 3 4 1.1 The source shall be sealed within the walls of a cylinder having a center hole capable of receiving a dosimeter of 9/16 inch in diameter The inside diameter of the cylinder shall not exceed 5/8 inch The location, intensity, and method of sealing this source shall be such that an IM-9( )/PD dosimeter inserted in the well for 4 minutes shall read 140 mr All seams in this unit shall be filled with silver solder so that no possibility of external contamination will exist The radiation exposure rate shall be no more than 40 mr/hr at any point on the external surface of the equipment

3 4 1.2 This cylinder shall be permanently fixed in the well of the cylinder.

3 4 2 Cylinder - The cylinder shall be made of cold rolled steel, unless otherwise approved by the command or agency concerned Dimensions for this cylinder shall not exceed 4-1/2 inches high and 3 inches in diameter

3 4 2.1 Well - The well shall be 3-3/4 inches in depth and the dimension of the well above the source shall be such as to accommodate a plug which shall offer the greatest resistance to any radiation leakage

3 4 3 Baseplate - A baseplate 1 inch greater in diameter than the chamber shall be welded to the bottom of the chamber to aid in the fabrication of a more stable piece of equipment

3 4 4 Carrying strap.- To facilitate carrying, the test chamber shall be provided with a carrying strap This strap shall be fabricated of a nonporous material designed to resist absorption of radioactive water or contaminated material.

3.4.5 Lock - A means shall be provided for locking the plug in place when the chamber is not in use This unit shall combine lock, strap, and plug

3 5 Range - The radiacmeter test chamber shall be capable of causing the 0 to 200 mr radiacmeter to read 140 mr per 4 minutes within the accuracy tolerance of the equipment

3 6 Accuracy - Any 0 to 200 mr or 0 to 1 R dosimeter in the Navy system inserted in the well of the test chamber for a predetermined period of time shall read within ±15 percent of the calibrated dosage

3 6.1 Calibration plate - A calibration plate shall be provided that will indicate the model number, predetermined time, and allowable limits of each type of 0 to 200 mr and 0 to 1 R dosimeter

3 7 Weight - The complete test chamber shall be as light as possible and, in any case, shall not exceed 15 pounds

3 8 Identification plate - An identification plate attached to the test chamber shall bear the following

(a) Warning of radiation
(b) Maximum field intensity at surface
(c) Nomenclature
(d) Unit number, strength, and calibration date of source

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3 8 1 The unit number, the exact strength of the source, the date of calibration, and half life of the source shall be inserted on the identification plate by the manufacturer calibrating the source.

3 9 External finish - The finish coat shall be gray enamel in accordance with class 2 of MIL-E-15090, and shall be applied as continuous films approximately 0 001 inch thick

4 QUALITY ASSURANCE PROVISIONS

4 1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements

4.2 Classification of inspection - The methods of examination and testing of the equipment fall within the following classifications:

(a) First article inspection
(b) Quality conformance inspection
Production inspection

4 3 General inspection - First article, production, and selected production models of the equipment shall pass the first article and quality conformance inspections specified herein and any other tests necessary, in the opinion of the cognizant inspector or the command or agency concerned, to determine compliance with this specification

4 4 First article inspection - First article inspection shall be made at a laboratory designated by the command or agency concerned on the first article model submitted by the contractor to determine suitability for Naval use First article inspection shall consist of any examination and tests necessary to determine conformance with this specification and shall include the following:

(a) Assembly, size, fit
(b) Type and character of material, parts, and finish, including prevention of corrosion.
(c) Operating test - the equipment shall be subjected to a practical operating test.

4 5 Quality conformance inspection.-

4.5 1 Production inspection - Production inspection shall be made at the place of manufacture by or under the supervision of the inspector on each equipment offered for delivery to determine compliance with this specification Production tests shall consist of the tests specified in 4 4

5 PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5 1 Packaging and packing - The equipment shall be packaged by level A or C and packed by level A, B, or C as specified in the contract or order in accordance with MIL-P-17555 Method III preservation shall apply for level A packaging As a minimum, the requirements shall conform to the Code of Federal Regulations, Title 49, governing the transportation of radioactive materials

5 2 Marking - The equipment and containers shall be marked in accordance with MIL-M-19590

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6 NOTES

6 1 Ordering data - Procurement documents should specify the following

(a) Title, number, and date of this specification
(b) Preservation, packaging, packing, or marking requirements other than those specified in 5 1

6 2 First article - Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement

6 3 Changes from previous issue - The outside margins of this document have been marked "#" to indicate where changes (deletions, additions, etc) from the previous issue have been made This has been done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written irrespective of the marginal notations and relationship to the last previous issue

Preparing activity
Navy - EC

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